

From Accuracy to Reliability: Risk-Aware Evaluation of Large Language Models for Decision-Support Under Sampling Variability

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Abstract

Large Language Models (LLMs) are increasingly used in decision-support tasks such as business analytics, planning, and automated reasoning systems. However, conventional evaluation methods focus primarily on accuracy, which fails to capture the reliability of model outputs under repeated sampling. A model may produce correct answers occasionally while still generating unstable or hallucinated reasoning that could mislead users in real-world applications.

This project investigates a risk-aware evaluation framework for LLM reasoning selection under sampling variability. Instead of generating a single answer, modern inference strategies often sample multiple reasoning chains for a given prompt. Our work studies how to select the most reliable reasoning path from a pool of sampled outputs. Using the monkey_business dataset from ScalingIntelligence, we compare three decision strategies: (1) raw single-sample selection, (2) majority voting across sampled answers, and (3) a lightweight verifier model that ranks candidate reasoning chains.

The verifier is implemented using a DistilBERT-based classifier trained to distinguish correct and incorrect reasoning traces. To evaluate system reliability beyond accuracy, we introduce a set of risk-aware metrics, including Decision Reliability Score (DRS), Decision Stability Index (DSI), Hallucination Rate (HR), Severe Hallucination Rate (SHR), and Risk Exposure Metric (REM).

Preliminary experiments demonstrate the feasibility of the evaluation pipeline, including dataset preprocessing, verifier training, and baseline comparison. The project aims to provide a practical framework for assessing LLM reasoning systems in settings where reliability, stability, and hallucination risk are as important as raw accuracy.

1 Introduction

Large Language Models (LLMs) have achieved impressive performance across many natural language processing tasks, including reasoning, question answering, and decision support. However, recent research has shown that LLM outputs can vary significantly across repeated sampling of reasoning chains. This variability raises concerns about reliability, particularly when models are used in business or analytical decision-support contexts.

Traditional evaluation methods focus primarily on single-output accuracy, which measures whether the model eventually produces a correct answer. While useful, this metric does not capture how stable or trustworthy the model's reasoning is across repeated samples. In real-world settings, users may rely on the first response produced by the system, making unstable reasoning or hallucinated explanations potentially harmful.

Recent studies have demonstrated that sampling multiple reasoning chains and selecting among them can improve performance. However, common aggregation methods such as majority voting assume that the most frequent answer is the most reliable. This assumption can fail when models repeatedly produce similar reasoning errors.

This project explores an alternative perspective: reliability-aware reasoning selection. Instead of focusing solely on whether a correct answer exists among generated samples, we investigate how reliably a system can select a trustworthy reasoning path from multiple candidates.

Our research addresses the following questions:

1. Can a lightweight verifier improve reasoning selection compared to single-sample outputs and majority voting?

2. How do risk-aware evaluation metrics change the ranking of decision strategies?
3. Can reliability signals such as hallucination risk and stability provide better evaluation of LLM reasoning systems?

2 Related Work

Recent research on large language models (LLMs) has examined several challenges related to reasoning reliability, including repeated sampling, verification-based evaluation, hallucination detection, and reliability-focused benchmarking. These studies highlight that traditional accuracy-based evaluation may not fully capture the reliability of model reasoning.

Repeated Sampling and Reasoning Aggregation

Recent work shows that generating multiple reasoning chains can improve reasoning performance. Brown et al. (2024) demonstrate that repeated sampling can reveal correct reasoning chains even when a single output is incorrect. Similarly, Taubenfeld et al. (2025) show that self-consistency methods can improve performance by aggregating multiple sampled outputs. However, these approaches typically rely on majority voting, which assumes that the most frequent answer is correct. This assumption may fail when models repeatedly produce similar reasoning errors. Our work addresses this limitation by studying verifier-based reasoning selection rather than relying only on frequency-based aggregation.

Verification and Error Detection

Another line of research focuses on verification mechanisms that evaluate candidate reasoning chains. Zhang et al. (2024) show that smaller language models can benefit from strong verifier models that evaluate generated reasoning. Similarly, Daheim et al. (2024) demonstrate that stepwise verification can help detect reasoning errors. However, Huang et al. (2024) show that language models often fail to correct their own mistakes without external verification. These findings motivate the use of independent verifier models to evaluate reasoning outputs.

Hallucination Detection and Reliability Evaluation

Several studies address hallucination detection and evaluation reliability in language models. Paudel et al. (2025) propose methods for detecting hallucinated outputs by verifying generated text against contextual knowledge. Lee et al. (2025) introduce a structured evaluation framework that improves the reliability of LLM evaluation through interpretable scoring criteria. Additionally, Dev et al. (2026) show that LLM-based judges can be unstable under small input changes, emphasizing the importance of robust evaluation frameworks.

Risk-Aware Evaluation of LLM Systems

Recent work also highlights the importance of risk-aware evaluation for real-world LLM applications. Nitsure et al. (2024) argue that benchmark accuracy alone is insufficient for evaluating deployed language model systems. Similarly, Nagaraja and Bahsi (2026) highlight the need for reliability evaluation in high-stakes decision-support settings.

Our work builds on these ideas by focusing on reasoning selection under sampling variability. In particular, our methodology draws inspiration from three studies that are closely related to our approach: Brown et al. (2024) on repeated sampling, Zhang et al. (2024) on verifier-based reasoning, and Lee et al. (2025) on structured evaluation frameworks. Following these works, our project compares raw selection, majority voting, and verifier-based ranking under a reliability-focused evaluation framework.

3 Methodology

3.1 Dataset and Task Definition

The project uses the ScalingIntelligence monkey_business dataset, which contains questions paired with multiple sampled reasoning chains generated by large language models. Each dataset instance includes: Question prompt, Multiple reasoning chains, Correctness labels, Ground truth answer. This dataset is particularly suitable because it allows direct analysis of sampling variability in reasoning outputs. The task is defined as a reasoning selection problem. Given a question and a set of candidate reasoning chains, the system must select the most reliable reasoning path.

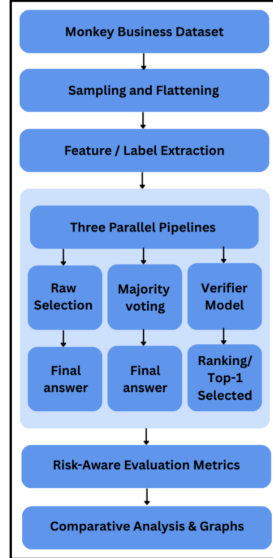
3.2 Data Preprocessing

The dataset is preprocessed using a balanced flattening approach. Each question is expanded into multiple question–reasoning pairs. Then correct and incorrect reasoning chains are labeled. After labeling a balanced subset of examples is created to avoid bias toward questions with many samples. Tokenization is performed using a transformer tokenizer with a maximum sequence length of 256 tokens.

3.3 Model Selection

The project uses a DistilBERT-based classifier as a lightweight verifier model. The verifier receives: Question + Reasoning Trace and predicts whether the reasoning chain is correct or incorrect. DistilBERT was chosen because it is computationally efficient and it provides strong performance on classification tasks. Also it enables scalable evaluation of large candidate sets.

3.4 System Pipeline



The system pipeline follows these steps:

1. Load dataset and extract reasoning samples
2. Preprocess and flatten reasoning traces
3. Train a verifier model
4. Apply different decision strategies

5. Evaluate outputs using risk-aware metrics

Decision strategies include:

- Raw selection: first generated reasoning chain
- Majority voting: most frequent answer
- Verifier ranking: select highest-scoring reasoning chain

3.5 Compute and Feasibility

Experiments are conducted using Python, Hugging Face Transformers, and PyTorch. Initial experiments are run using subset-based evaluation on GPU-enabled environments such as Google Colab.

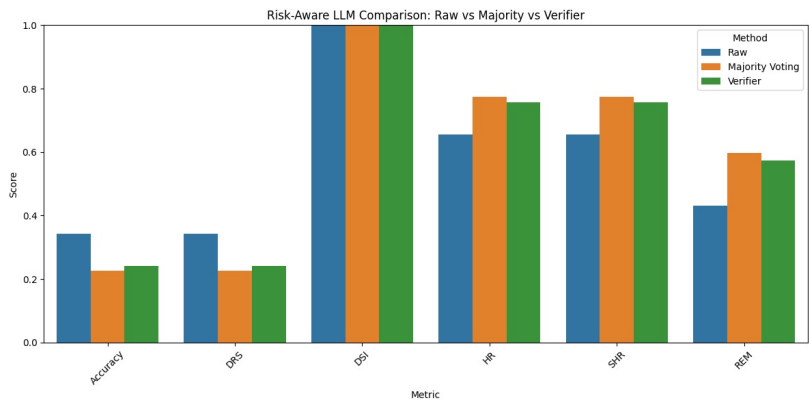
The lightweight verifier model enables efficient evaluation without requiring large-scale infrastructure.

4 Evaluation Framework

The evaluation framework includes both traditional and risk-aware metrics.

Category	Metric	Definition
Standard Metric	Accuracy	Proportion of correct selected answers.
Reliability Metric	Decision Reliability Score (DRS)	Measures how consistently the system selects correct answers.
	Decision Stability Index (DSI)	Measures variability in model decisions across samples.
Risk Metric	Hallucination Rate (HR)	Proportion of outputs containing unsupported reasoning.
	Severe Hallucination Rate (SHR)	Rate of critical reasoning failures.
	Risk Exposure Metric (REM)	Aggregated risk score across outputs.

These metrics allow evaluation of both correctness and reliability.



5 Expected Contributions

This project makes several contributions:

1. A risk-aware evaluation framework for LLM reasoning systems.
2. Analysis of sampling variability in reasoning outputs.
3. Comparison of three reasoning selection strategies.
4. Demonstration of a lightweight verifier model for reliable reasoning selection.

These contributions aim to improve the evaluation of LLMs used in real-world decision-support systems.

6 Progress and Next Steps

Component	Status	Details
Dataset Pipeline	Complete	Loaded samples, ground truth, labels
Data Preprocessing	Complete	Balanced flattening, equal sampling per question
Decision Strategies	Complete	Raw, Majority, Verifier implemented
Metrics Framework	Complete	Accuracy, DRS, DSI computed
Risk Evaluation	Complete	HR, SHR, REM implemented
Comparative Analysis	Complete	Graphs generated, trade-offs identified
Dataset Expansion	Planned	Integrate GSM8K and CodeContests
Verifier Improvement	Planned	Improve training and ranking
Final Paper	Planned	Results, discussion, conclusion

We have expanded the project to include three datasets, and for each dataset we are evaluating three decision methods: baseline model output, majority voting, and verifier-based selection. We are currently implementing comparative visualizations that show method-wise performance within each dataset, pairwise comparisons between datasets, and an overall three-dataset comparison. This will allow us to analyze differences in accuracy, reliability, and risk-aware behavior across both methods and datasets.

7 Conclusion

This project investigates how to evaluate LLM reasoning systems beyond accuracy by focusing on reliability and risk. By combining repeated sampling, verifier-based selection, and risk-aware metrics, the proposed framework aims to provide a more realistic evaluation of LLM decision-support systems. Preliminary progress demonstrates the feasibility of the pipeline and sets the stage for larger-scale experiments in the final phase of the project.

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